

Math 162

Final Exam solutions

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1. (a) Clearly f is continuous on $(0, 1]$. We can use L'Hôpital's rule to compute

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = \lim_{x \rightarrow 0} \frac{\cos x}{1} = 1,$$

which implies that f is continuous at 0 as well. Thus f is continuous on $[0, 1]$, hence integrable.

- (b) Since $\cos 0 = 1$, we have

$$\lim_{x \rightarrow 0^+} \frac{\cos x}{x} = \infty.$$

This shows that f is unbounded on $[0, 1]$ and hence not integrable.

- (c) Observe that

$$\lim_{x \rightarrow 0^+} \frac{\log x}{x} = -\infty,$$

since $\log x$ approaches $-\infty$ and $\frac{1}{x}$ approaches ∞ as $x \rightarrow 0^+$. So f is unbounded on $[0, 1]$ and therefore not integrable.

2. The first fundamental theorem of calculus and the chain rule imply that

$$\frac{d}{dx} \int_0^{g(x)} h(t) dt = h(g(x))g'(x),$$

and similarly we have

$$\frac{d}{dx} \int_0^{f(x)} h(t) dt = h(f(x))f'(x).$$

Combining these with the standard properties of the integral, we obtain

$$\frac{d}{dx} \int_{f(x)}^{g(x)} h(t) dt = \frac{d}{dx} \int_0^{g(x)} h(t) dt - \frac{d}{dx} \int_0^{f(x)} h(t) dt = h(g(x))g'(x) - h(f(x))f'(x).$$

3. (a) For any $N \in \mathbb{N}$, we have

$$\sum_{i=1}^N |a_{n_i}| \leq \sum_{k=1}^{n_N} |a_k| \leq \sum_{k=1}^{\infty} |a_k|,$$

where the first inequality holds because the first sum is just the second sum with some terms omitted. It follows that the (nonnegative) sequence of partial sums

$$\left(\sum_{i=1}^N |a_{n_i}| \right)_{N \in \mathbb{N}}$$

is bounded above, hence convergent. This means that $\sum_{i=1}^{\infty} a_{n_i}$ converges absolutely and therefore converges.

(b) Consider the alternating harmonic series

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots,$$

which we have seen converges conditionally. The series

$$-\frac{1}{2} - \frac{1}{4} - \frac{1}{6} - \cdots = -\frac{1}{2} \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots \right)$$

consisting of its evenly numbered terms diverges, being a constant multiple of the harmonic series.

4. Recall the Taylor series expansion

$$\cos x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!},$$

valid for all $x \in \mathbb{R}$. Substitute $3x^5$ into this series to obtain

$$\cos(3x^5) = \sum_{n=0}^{\infty} (-1)^n \frac{3^{2n} x^{10n}}{(2n)!}.$$

Now subtract 1 and divide by x^7 , which yields

$$f(x) = \sum_{n=1}^{\infty} (-1)^n \frac{3^{2n} x^{10n-7}}{(2n)!}$$

(one has to assume that $x \neq 0$ to divide by x^7 , but note that both sides of the equation vanish when $x = 0$, so it is valid in that case as well).

The formula for Taylor coefficients therefore implies that

$$\frac{f^{(10n-7)}(0)}{(10n-7)!} = (-1)^n \frac{3^{2n}}{(2n)!}$$

for any $n \in \mathbb{N}$. Since $2023 = 10 \cdot 203 - 7$, it follows that

$$f^{(2023)}(0) = 2023! \cdot (-1)^{203} \frac{3^{2 \cdot 203}}{(2 \cdot 203)!} = -\frac{2023! \cdot 3^{406}}{406!}.$$

5. We recall that the Taylor expansion of $\arctan x$ at $x = 0$ can be derived as follows. We have the geometric series formula

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots,$$

valid for $|x| < 1$. Since power series can be integrated term by term inside their open interval of convergence, we obtain

$$\arctan x = \int_0^x \frac{dt}{1+t^2} = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots,$$

again for $|x| < 1$.

We would like to evaluate this identity at $x = 1$, but so far we have not proved its validity in that case. For this, notice that for any $n \in \mathbb{N}$, the identity

$$\frac{1}{1+x^2} = 1 - x^2 + \cdots + (-1)^{n-1} x^{2n-2} + (-1)^n \frac{x^{2n}}{1+x^2}$$

holds for all $x \in \mathbb{R}$ (for example, this can be seen by multiplying both sides by $1+x^2 \neq 0$). Integrating, we obtain

$$\arctan x = x - \frac{x^3}{3} + \cdots + (-1)^{n-1} \frac{x^{2n-1}}{2n-1} + (-1)^n \int_0^x \frac{t^{2n}}{1+t^2} dt,$$

again valid for all $x \in \mathbb{R}$. Now observe that

$$0 \leq \int_0^1 \frac{t^{2n}}{1+t^2} dt \leq \int_0^1 t^{2n} dt = \frac{1}{2n+1},$$

so by the squeeze theorem we obtain

$$\lim_{n \rightarrow \infty} \int_0^1 \frac{t^{2n}}{1+t^2} dt = 0.$$

Since

$$\frac{\pi}{4} = \arctan 1 = 1 - \frac{1}{3} + \cdots + (-1)^{n-1} \frac{1}{2n-1} + (-1)^n \int_0^1 \frac{t^{2n}}{1+t^2} dt,$$

upon passing to the limit as $n \rightarrow \infty$ it follows that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$$

as desired.